Brain

How do you remember the way to your friend's house? Why do your eyes blink without you ever thinking about it? Where do dreams come from? Your brain is in charge of these things and a lot more.

In fact, your brain is the boss of your body. It runs the show and controls just about everything you do, even when you're asleep. Not bad for something that looks like a big, wrinkly, gray sponge.

Your brain has many different parts that work together. We're going to talk about these five parts, which are key players on the brain team:

cerebrum (say: suh-ree-brum)

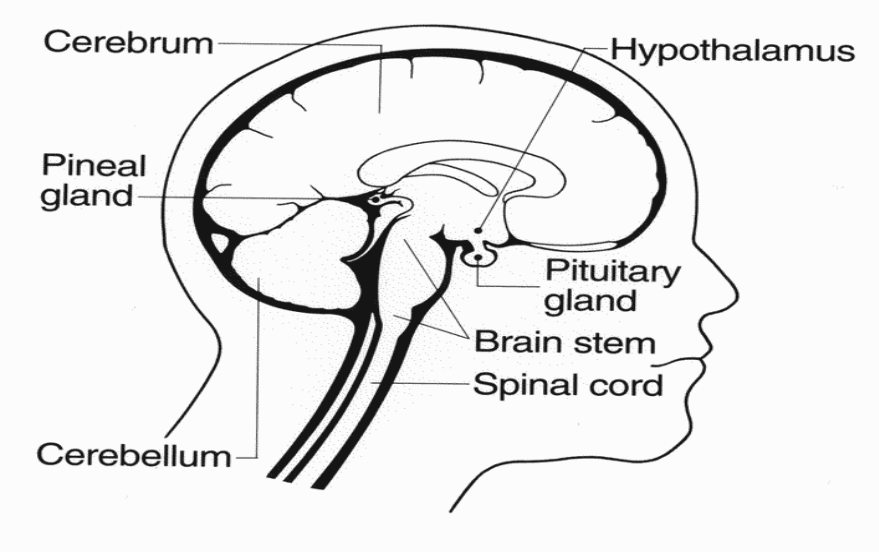
cerebellum (say: sair-uh-bell-um)

brain stem

pituitary gland (say: puh-too-uh-ter-ee gland)

hypothalamus (say: hy-po-thal-uh-mus)

The Biggest Part: the Cerebrum. Colour just the cerebrum in on the diagram below.



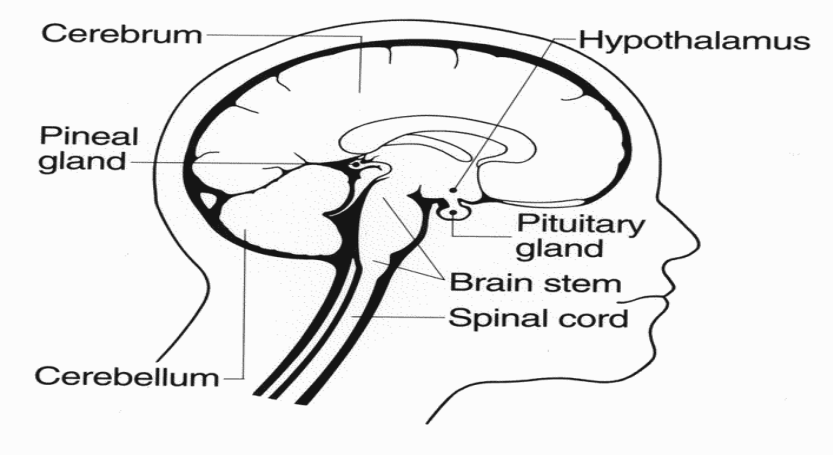
The biggest part of the brain is the cerebrum. The cerebrum makes up 85% of the brain's weight, and it's easy to see why. The cerebrum is the thinking part of the brain and it controls your voluntary muscles — the ones that

move when you want them to. So you can't dance — or kick a soccer ball — without your cerebrum.

When you're thinking hard, you're using your cerebrum. You need it to solve math problems, figure out a video game, and draw a picture. Your memory lives in the cerebrum — both short-term memory (what you ate for dinner last night) and long-term memory (the name of that roller-coaster you rode on two summers ago). The cerebrum also helps you reason, like when you figure out that you'd better do your homework now because your mom is taking you to a movie later.

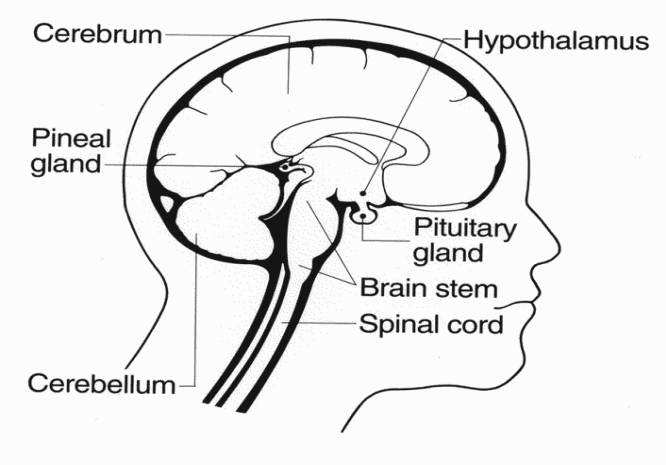
The cerebrum has two halves, with one on either side of the head. Some scientists think that the right half helps you think about abstract things like music, colors, and shapes. The left half is said to be more analytical, helping you with math, logic, and speech. Scientists do know for sure that the right half of the cerebrum controls the left side of your body, and the left half controls the right side.

The Cerebellum's Balancing Act. Colour in the Cerebellum on the diagram below.'



Next up is the cerebellum. The cerebellum is at the back of the brain, below the cerebrum. It's a lot smaller than the cerebrum at only 1/8 of its size. But it's a very important part of the brain. It controls balance, movement, and coordination (how your muscles work together). Because of your cerebellum, you can stand upright, keep your balance, and move around. Think about a surfer riding the waves on his board. What does he need most to stay balanced? The best surfboard? The coolest wetsuit? Nope — he needs his cerebellum!

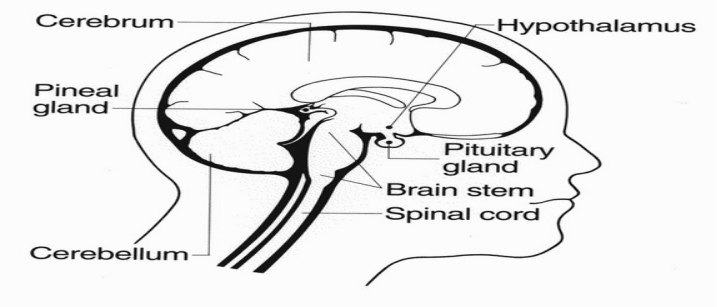
Brain Stem Keeps You Breathing — and More. Colour in the brain stem in the diagram below.



Another brain part that's small but mighty is the brain stem. The brain stem sits beneath the cerebrum and in front of the cerebellum. It connects the rest of the brain to the spinal cord, which runs down your neck and back. The brain stem is in charge of all the functions your body needs to stay alive, like breathing air, digesting food, and circulating blood.

Part of the brain stem's job is to control your involuntary muscles — the ones that work automatically, without you even thinking about it. There are involuntary muscles in the heart and stomach, and it's the brain stem that tells your heart to pump more blood when you're biking or your stomach to start digesting your lunch. The brain stem also sorts through the millions of messages that the brain and the rest of the body send back and forth. Whew! It's a big job being the brain's secretary!

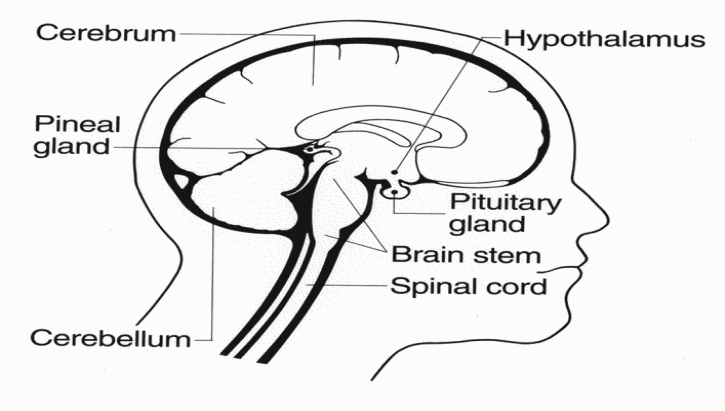
Pituitary Gland Controls Growth and other things. Colour in the pituitary in the diagram below.



The pituitary gland is very small — only about the size of a pea! Its job is to produce and release hormones into your body. If your clothes from last year are too small, it's because your pituitary gland released special hormones that made you grow. This gland is a big player in puberty too. This is the time when boys' and girls' bodies go through major changes as they slowly become men and women, all thanks to hormones released by the pituitary gland.

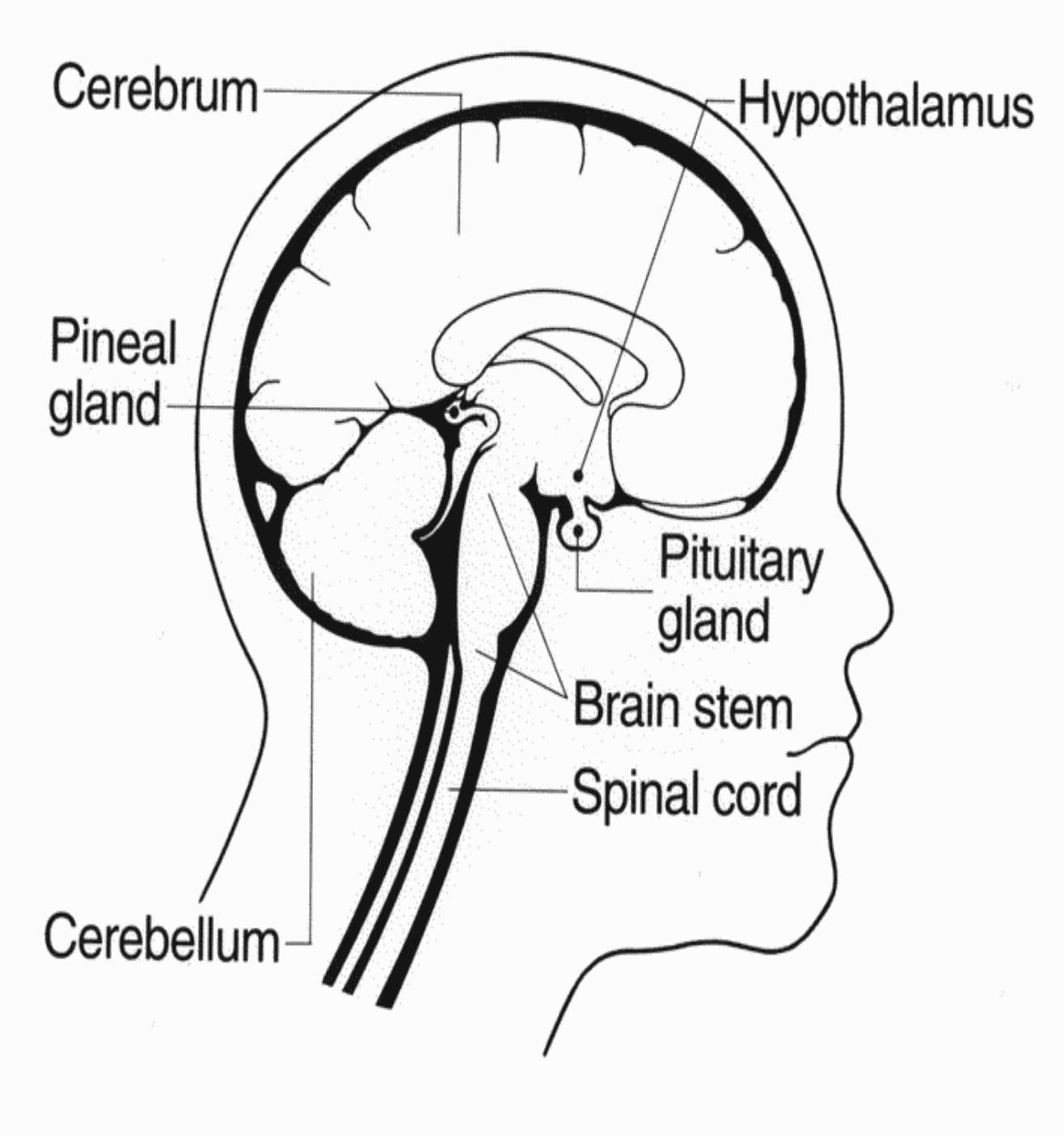
This little gland also plays a role with lots of other hormones, like ones that control the amount of sugars and water in your body. And it helps keep your metabolism (say: muh-ta-buh-lih-zum) going. Your metabolism is everything that goes on in your body to keep it alive and growing and supplied with energy, like breathing, digesting food, and moving your blood around.

Hypothalamus Controls Temperature and other things. Colour in the hypothalamus in the diagram below.

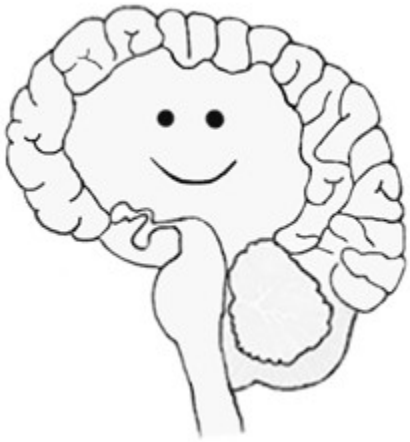


The hypothalamus is like your brain's inner thermostat (that little box on the wall that controls the heat in your house). The hypothalamus knows what temperature your body should be (about 98.6° Fahrenheit or 37° Celsius). If your body is too hot, the hypothalamus tells it to sweat. If you're too cold, the hypothalamus gets you shivering. Both shivering and sweating are attempts to get your body's temperature back where it needs to be.

Look at each part of the brain shown below. Write onto that part of the brain some words that describe the jobs that part of the brain does.



Be Good to Your Brain



Eat healthy foods. They contain proteins, potassium and calcium (minerals that are important for the brain). Get a lot of playtime (exercise). Wear a helmet when you ride your bike or play other sports that require head protection. Wear a helmet in areas where heavy machinery or construction is occurring. You do not want something to fall on your head and squish your brain. Don't drink too much alcohol, take drugs, or use tobacco. Use your brain by doing challenging activities, such as puzzles, reading, playing music, making art, or anything else that gives your brain a workout! Get a good night’s sleep. Avoid working around dangerous chemicals that contain heavy metals, like lead and mercury. Avoid working around dangerous solvents(some of these solvents are used in glues and paints)

Brain poster. Choose one of the things listed in the be good to your brain Box above and make up a simple poster with a slogan and picture that helps advertise it.